Ministerul Educației și Cercetării al Republicii Moldova
Universitatea Tehnică a Moldovei
Facultatea Calculatoare, Informatică și Microelectronică

Raport

la

Lucrarea de laborator Nr. 7

Disciplina: Tehnici de programare

Tema: „GitHub, pas cu pas”
Lucrarea A

A efectuat: st. gr. SI-264 Danuța Daniel

A verificat: asistent universitar Reițman Patricia

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SARCINA LUCRĂRII

Exersarea comenzilor Git pentru clonare, commit, sincronizare și verificarea istoricului.

Mersul lucrării

A1-A2

```
danu1@Danus-MacBook-Air ~ % git --version
git version 2.50.1 (Apple Git-155)
danu1@Danus-MacBook-Air ~ % git config --global user.name
danuv98
danu1@Danus-MacBook-Air ~ %
```

Fig. 1.1 – Contul GitHub

```
danu1@Danus-MacBook-Air ~ % git config --global init.defaultBranch main
danu1@Danus-MacBook-Air ~ % git config --global pull.rebase false
danu1@Danus-MacBook-Air ~ % git config --global --list
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
user.name=danuv98
user.email=[redacted]
credential.helper=cache
init.defaultbranch=main
pull.rebase=false
danu1@Danus-MacBook-Air ~ %
```

Fig. 1.2 – Configurarea Git

A3

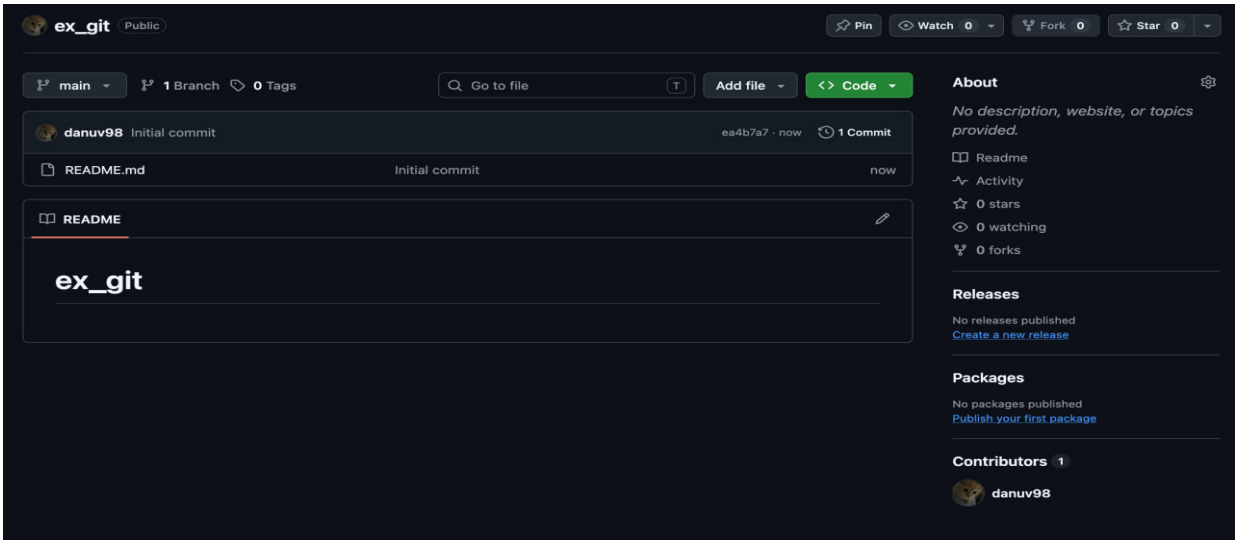


Fig. 2.1 – Depozitul utilizat

A4

```
danu1@Danus-MacBook-Air Laborator_8 % git clone https://github.com/danuv98/ex_git.git

Cloning into 'ex_git'...
remote: Enumerating objects: 3, done.
remote: Counting objects: 100% (3/3), done.
remote: Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Receiving objects: 100% (3/3), done.
danu1@Danus-MacBook-Air Laborator_8 % cd ex_git
danu1@Danus-MacBook-Air ex_git % ls
README.md
danu1@Danus-MacBook-Air ex_git %
```

Fig. 2.2 – Clonarea depozitului

A5

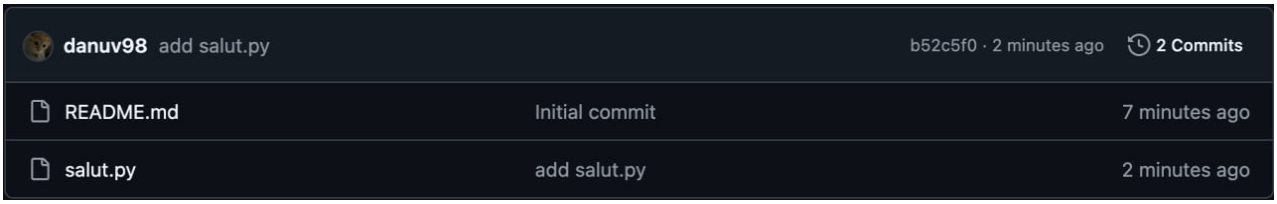
```
danu1@Danus-MacBook-Air ex_git % printf 'print("Salut, GitHub!")\n' > salut.py
danu1@Danus-MacBook-Air ex_git % python3 salut.py
Salut, GitHub!
danu1@Danus-MacBook-Air ex_git % git status
On branch main
Your branch is up to date with 'origin/main'.

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        salut.py

nothing added to commit but untracked files present (use "git add" to track)
```

Fig. 3.1 – Status add și commit

A6




```
danu1@Danus-MacBook-Air ex_git % git push
Enumerating objects: 4, done.
Counting objects: 100% (4/4), done.
Delta compression using up to 8 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 300 bytes | 300.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/danuv98/ex_git.git
   ea4b7a7..b52c5f0  main -> main
```

Fig. 3.2 – Publicarea ramurii

A7

```
danu1@Danus-MacBook-Air ex_git % git pull

Already up to date.
danu1@Danus-MacBook-Air ex_git % cat README.md
# ex_git

aici exersez git
danu1@Danus-MacBook-Air ex_git %
```

Fig. 4.1 – Sincronizarea prin pull

A8

```
danu1@Danus-MacBook-Air ex_git % python3 -m venv .venv

danu1@Danus-MacBook-Air ex_git % git status --short
?? .venv/
danu1@Danus-MacBook-Air ex_git % printf '.venv/\n__pycache__/\n' > .gitignore
danu1@Danus-MacBook-Air ex_git % cat .gitignore
.venv/
__pycache__/
danu1@Danus-MacBook-Air ex_git % git status --short
?? .gitignore
danu1@Danus-MacBook-Air ex_git % git check-ignore .venv/pyvenv.cfg
.venv/pyvenv.cfg
danu1@Danus-MacBook-Air ex_git %
```

Fig. 4.2 – Ignorarea mediului virtual

A9

```
danu1@Danus-MacBook-Air ex_git % git log --oneline
cc415a0 (HEAD -> main, origin/main, origin/HEAD) Ignor mediul virtual
917a0b1 Update README.md
b52c5f0 add salut.py
ea4b7a7 Initial commit
danu1@Danus-MacBook-Air ex_git % printf 'print("al doilea")\n' >> salut.py
danu1@Danus-MacBook-Air ex_git % git diff
diff --git a/salut.py b/salut.py
index e700475..b7c6a14 100644
--- a/salut.py
+++ b/salut.py
@@ -1 +1,2 @@
 print("Salut, GitHub!")
+print("al doilea")
```

```
danu1@Danus-MacBook-Air ex_git % git restore salut.py
danu1@Danus-MacBook-Air ex_git % git status
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
danu1@Danus-MacBook-Air ex_git %
```

Fig. 5.1 – Log diff și restore

A10

```
danu1@Danus-MacBook-Air ex_git % cd ~/Desktop/Laborator_8
danu1@Danus-MacBook-Air Laborator_8 % mkdir exercitiu-2
danu1@Danus-MacBook-Air Laborator_8 % cd exercitiu-2
danu1@Danus-MacBook-Air exercitiu-2 % git init
Initialized empty Git repository in /Users/danu1/Desktop/Laborator_8/exercitiu-2/.git/
danu1@Danus-MacBook-Air exercitiu-2 % printf '# second repo\n' > README.md
danu1@Danus-MacBook-Air exercitiu-2 % git add README.md
danu1@Danus-MacBook-Air exercitiu-2 % git commit -m "primul commit"
[main (root-commit) a158a93] primul commit
 1 file changed, 1 insertion(+)
 create mode 100644 README.md
danu1@Danus-MacBook-Air exercitiu-2 %
```

```
danu1@Danus-MacBook-Air exercitiu-2 % git branch -M main
danu1@Danus-MacBook-Air exercitiu-2 % git push -u origin main
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 224 bytes | 224.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To https://github.com/danuv98/ex_git2.git
 * [new branch]      main -> main
branch 'main' set up to track 'origin/main'.
danu1@Danus-MacBook-Air exercitiu-2 % git remote add origin https://github.com/danuv98/ex_git2.git
error: remote origin already exists.
danu1@Danus-MacBook-Air exercitiu-2 %
```

Fig. 5.2 – Demonstrația locală git init

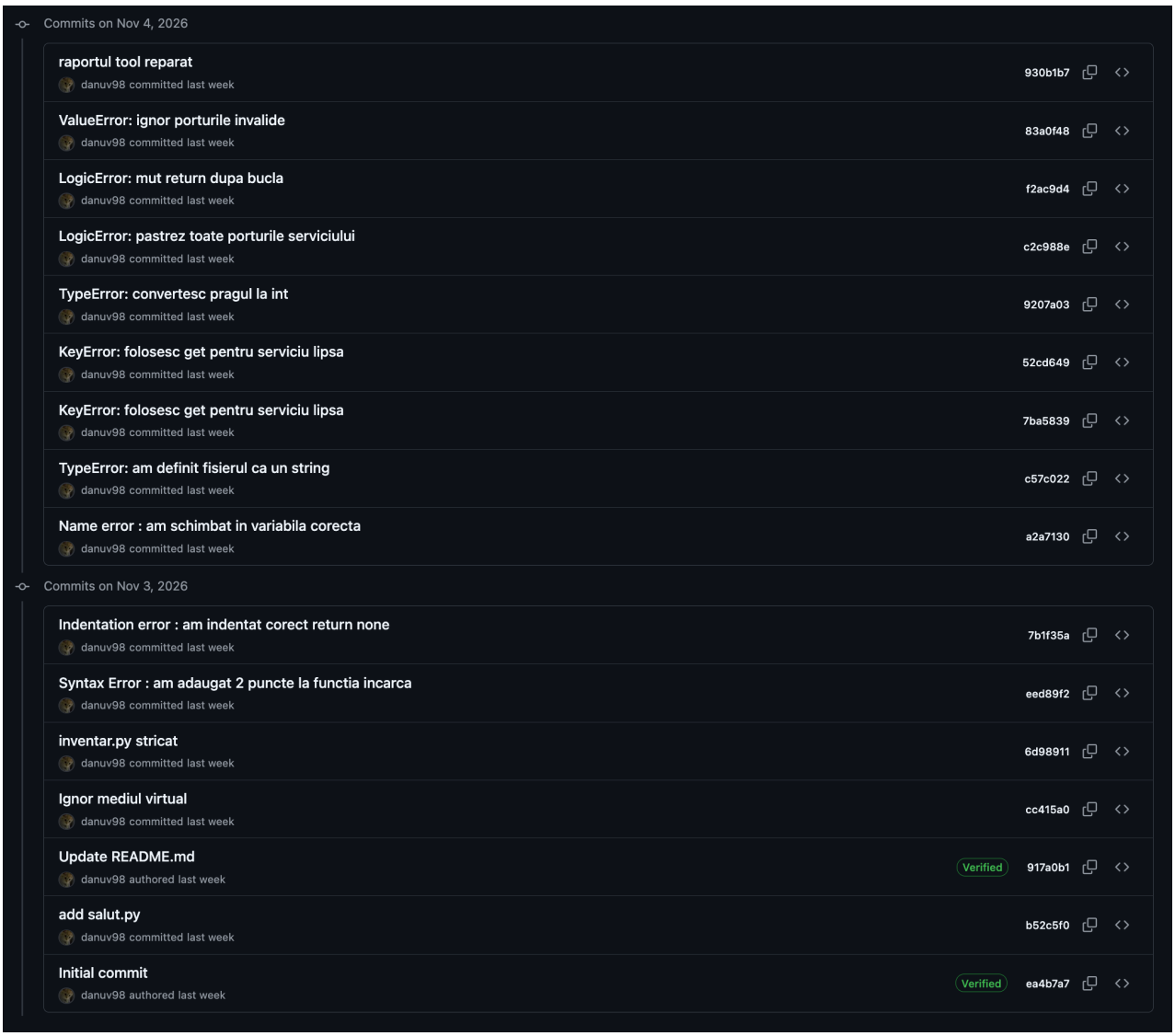


Fig 6.1 – Commit History

```python
# inventar.py - rezumatul tintelor dintr-un fisier JSON
# Unealta a fost scrisa in graba si nu mai merge. Reparati-o.
# Rulare:  python inventar.py tinte.json --prag 1000
import argparse
import json


def incarca(cale):
    """Citeste lista de tinte din fisierul JSON."""
    with open(cale) as f:
        return json.load(f)


def port_valid(text):
    """Intoarce portul ca int, sau None daca nu e un numar."""
    try:
        return int(text)
    except ValueError:
        return None


def pe_servicii(tinte):
    """Dict: serviciu -> lista de porturi."""
    grupe = {}
    for t in tinte:
        serviciu = t.get("serviciu", "necunoscut")
        port = port_valid(t["port"])
        if port is None:
            print("sar peste", t["gazda"], ": port invalid", t["port"])
            continue
        if serviciu not in grupe:
            grupe[serviciu] = []
        grupe[serviciu].append(port)

    return grupe


def peste_prag(tinte, prag):
    """Lista gazdelor cu un port mai mare decat pragul."""
    rezultat = []
    for t in tinte:
        port = port_valid(t["port"])
        if port is None:
            continue
        if port > prag:
            rezultat.append(t["gazda"])

    return rezultat


def main():
    p = argparse.ArgumentParser()
    p.add_argument("fisier")
    p.add_argument("--prag", type=int, default=1024)
    a = p.parse_args()

    tinte = incarca(a.fisier)
    print("Tinte citite: " + str(len(tinte)))

    grupe = pe_servicii(tinte)
    for serviciu, porturi in grupe.items():
        print(f"{serviciu}: {porturi}")

    mari = peste_prag(tinte, a.prag)
    print(f"Peste {a.prag}: {mari}")

    with open("raport_inventar.json", "w") as f:
        json.dump({"servicii": grupe, "peste_prag": mari}, f, indent=2)
    print("Raport salvat in raport_inventar.json")


if __name__ == "__main__":
    main()
```

Fig 6.2 – Cod Final

Concluzii

Lucrarea urmărește exersarea commit-urilor, sincronizarea cu GitHub și inițializarea unui depozit local.